

Problem Description

We need to specify the requirements of a new system called HomeSphere that simplifies the process of listing, searching, and managing residential rental properties. The platform will primarily focus on apartments, houses, and single-room rentals.

Currently, property owners manage rental information manually through phone calls, spreadsheets, and social media advertisements. This process is inefficient, difficult to maintain, and often results in outdated or incomplete property information. Customers also face challenges in finding suitable rental properties because there is no centralized system for browsing and comparing available listings. HomeSphere intends to solve these problems by creating a centralized online platform that streamlines communication between customers and property owners while improving the overall rental experience.

The system must be simple, intuitive, and user-friendly. Customers should be able to browse available properties by category and location. For example, users should be able to select categories such as “Apartments,” “Houses,” or “Single Rooms,” and further refine search results using filters such as city, rental price range, number of bedrooms, and property availability.

When a customer selects a property, the system should display detailed information about the listing, including:

- Property title and description
- Rental price
- Address and location
- Number of bedrooms and bathrooms
- Property images
- Availability status
- Contact information for the property owner

The platform should support customer account registration and authentication. Registered customers should be able to store personal information securely, manage their profiles, and save favorite properties for future reference. Customers should also be able to send rental inquiries, request appointments for property visits, and receive responses from property owners through the platform.

Property owners should be able to create accounts and manage their own listings. Owners must be able to:

- Add new property listings
- Upload and manage property images
- Edit property details and pricing
- Mark properties as rented or unavailable
- Remove outdated listings

The system should also facilitate communication between customers and property owners. Customers should be able to send messages or appointment requests directly through the platform, while property owners should receive notifications through email or the application dashboard.

In addition, the application should include an administrative section for platform staff. Administrators should be able to:

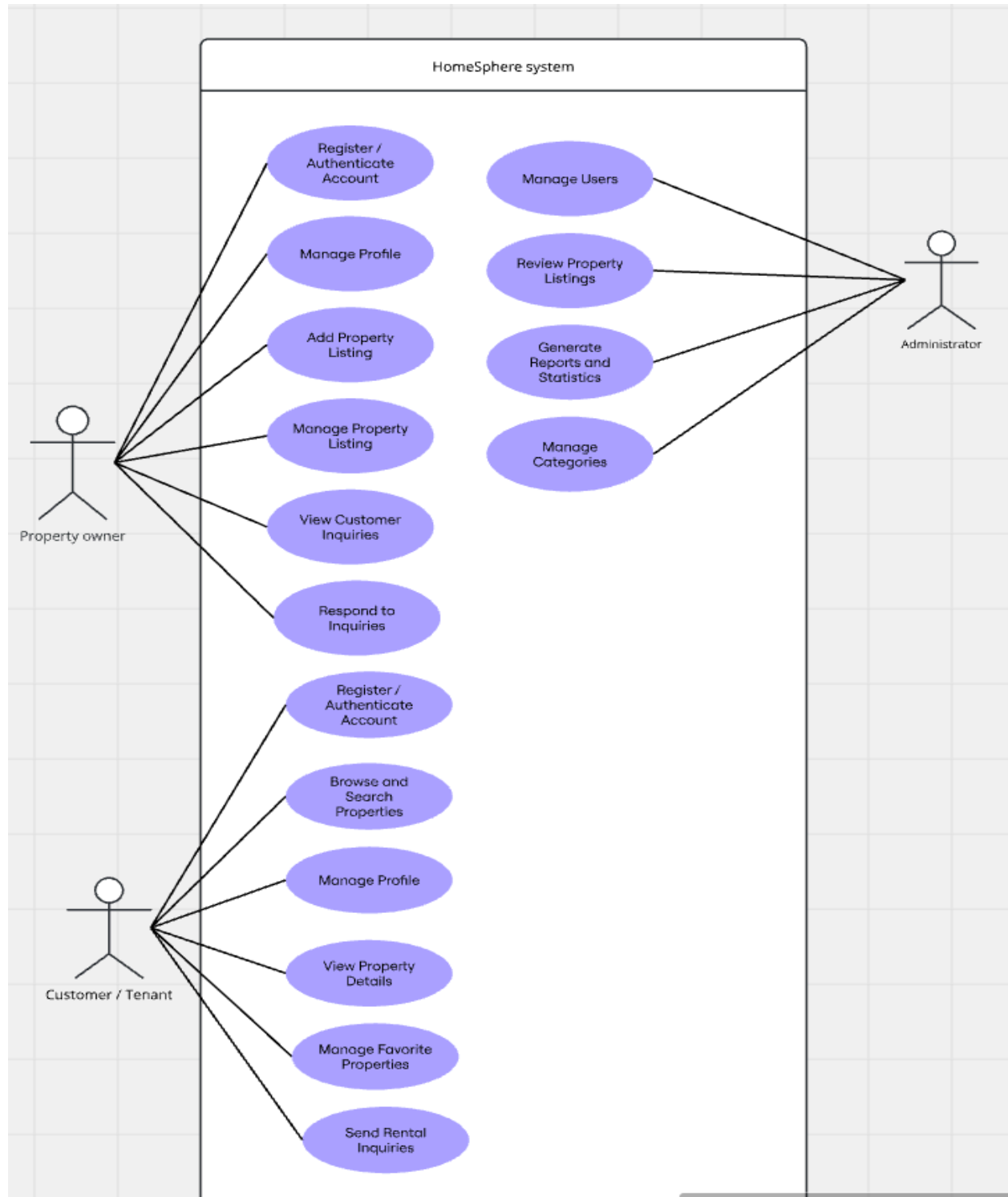
- Manage customer and owner accounts
- Review and approve property listings
- Remove inappropriate, duplicate, or outdated content
- Monitor platform activity

The system should also generate management reports and statistics to support business decisions. Example reports may include:

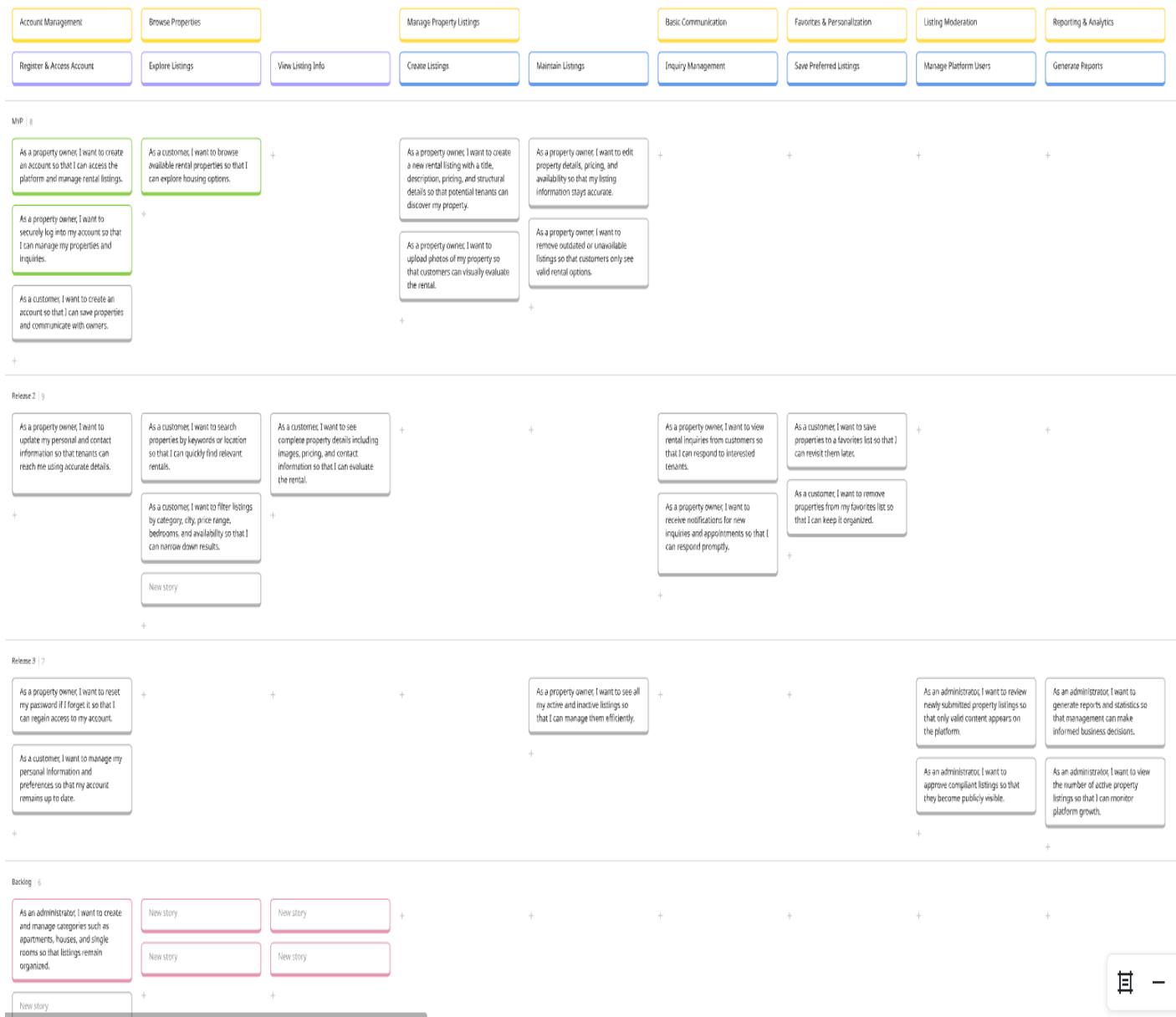
- Number of active property listings
- Number of registered users
- Number of rental inquiries
- Most viewed properties
- Listings by category or location

The HomeSphere platform is expected to improve efficiency for property owners, provide a better search experience for customers, and create a centralized, reliable, and scalable rental management solution.

Use-Case Diagram



User Story Map



Acceptance Criteria

User Story 1: Customer Search & Filter

As a customer, I want to filter property listings by city, price range, and category so that I can quickly find a home that matches my budget and location preferences.

Acceptance Criteria

- Customers can filter properties by category, city, and price range simultaneously
- System updates search results in real-time when filters are applied
- The system displays a clear "No matching properties found" message if no listings meet the criteria
- The customer can reset all active filters with a single click back to the default view

User Story 2: Owner Adding a Listing

As a property owner, I want to create a new rental listing with a title, description, pricing, and structural details so that potential tenants can discover my property.

Acceptance Criteria

- The owner can create a new listing by filling out a property details form
- The owner can't submit a listing with missing required fields (Title, Price, Address, Category)
- The owner can't input negative values or text into the price and room count fields
- System saves newly submitted properties under a "Pending Approval" status before publishing

User Story 3: Admin Reviewing & Approving Listings

As a platform administrator, I want to review newly submitted property listings so that I can approve compliant posts and prevent fraudulent or inappropriate content from going live.

Acceptance Criteria

- Admin can view a dedicated queue of all "Pending Approval" property listings
- Admin can change a listing status to "Active", making it immediately visible in public searches
- Admin can't approve a listing without reviewing its full details page
- Admin can reject a listing and must select or provide a rejection reason

Main Scenarios

User Story 1

1. The customer navigates to the HomeSphere search page.
2. The system fetches and displays all active property listings by default.
3. The customer clicks the "Category" dropdown and selects "Apartment".
4. The customer types a specific city into the search box.
5. The customer adjusts the minimum and maximum boundary sliders on the price range filter.
6. The customer clicks the "Apply Filters" button.
7. The system filters the database in real-time and refreshes the page to show only matching apartments.

User Story 2

1. The property owner logs into their HomeSphere account.
2. The owner clicks the "Add New Listing" button on their dashboard.
3. The system renders the property details creation form.
4. The owner enters the property title, detailed description, address, images, monthly rent price, and drops down to select the number of bedrooms and bathrooms.
5. The owner hits the "Submit for Review" button.
6. The system validates the data formats and completeness.
7. The system saves the property state as "Pending Approval" and redirects the owner back to their dashboard with a banner indicating success.

User Story 3

1. The administrator logs into the secure HomeSphere administrative portal.
2. The admin navigates to the "Content Moderation Queue" menu.
3. The system displays a list of all property submissions currently flagged as "Pending Approval".
4. The admin clicks on a specific listing to open the full detailed draft review page.
5. The admin scans the text, address, and pricing parameters for policy compliance.

6. The admin clicks the green "Approve Listing" button.
7. The system updates the listing status to "Active" in the database, indexes it for customer search results, and logs the admin ID responsible for the approval.